Aerial Survey ROI Workflow — User Guide

This guide walks you through adding, configuring, and running the Aerial Survey ROI Workflow, which generates aerial survey transect lines over a user-defined Region of Interest (ROI).

> **Note:** This workflow only supports **polygon** geometry. Point and line geometry types are not supported.

Overview

The workflow produces:

- An interactive HTML ecomap showing the survey lines overlaid on the ROI
- Survey lines exported as a GeoParquet (`.geoparquet`)
- A dashboard widget displaying the map

Prerequisites

Before running the workflow, ensure you have:

- Access to an Ecoscope Workflows deployment (UI or API)
- A geospatial file defining your Region of Interest — GeoPackage (`.gpkg`), GeoJSON, or GeoParquet (`.geoparquet`). **The file must contain polygon geometry.**
- The `ECOSCOPE_WORKFLOWS_RESULTS` environment variable set to a writable output directory on your deployment

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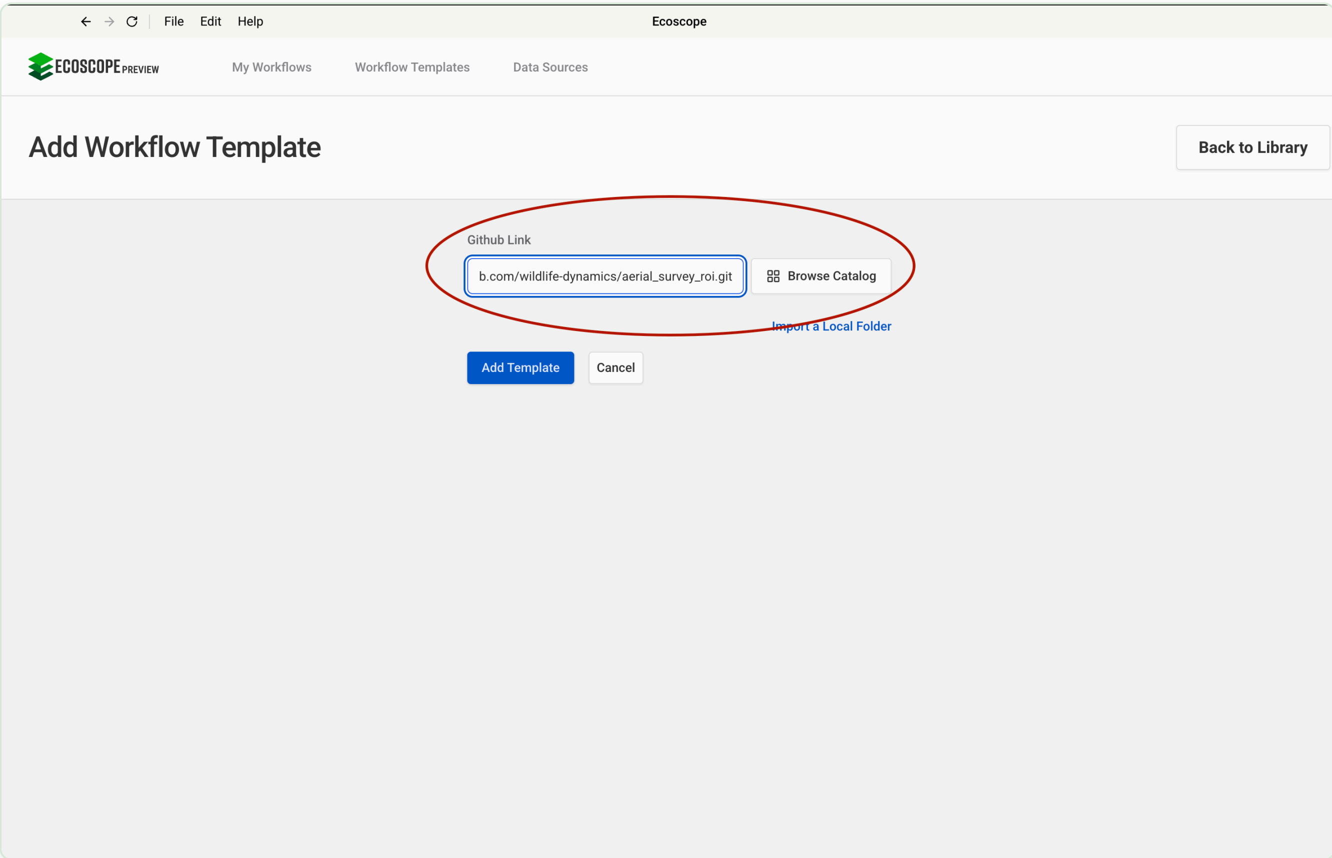
Step-by-Step Configuration

Step 1 — Add the Workflow Template

Navigate to **Workflow Templates** in the top navigation bar and click **Add Workflow Template**.

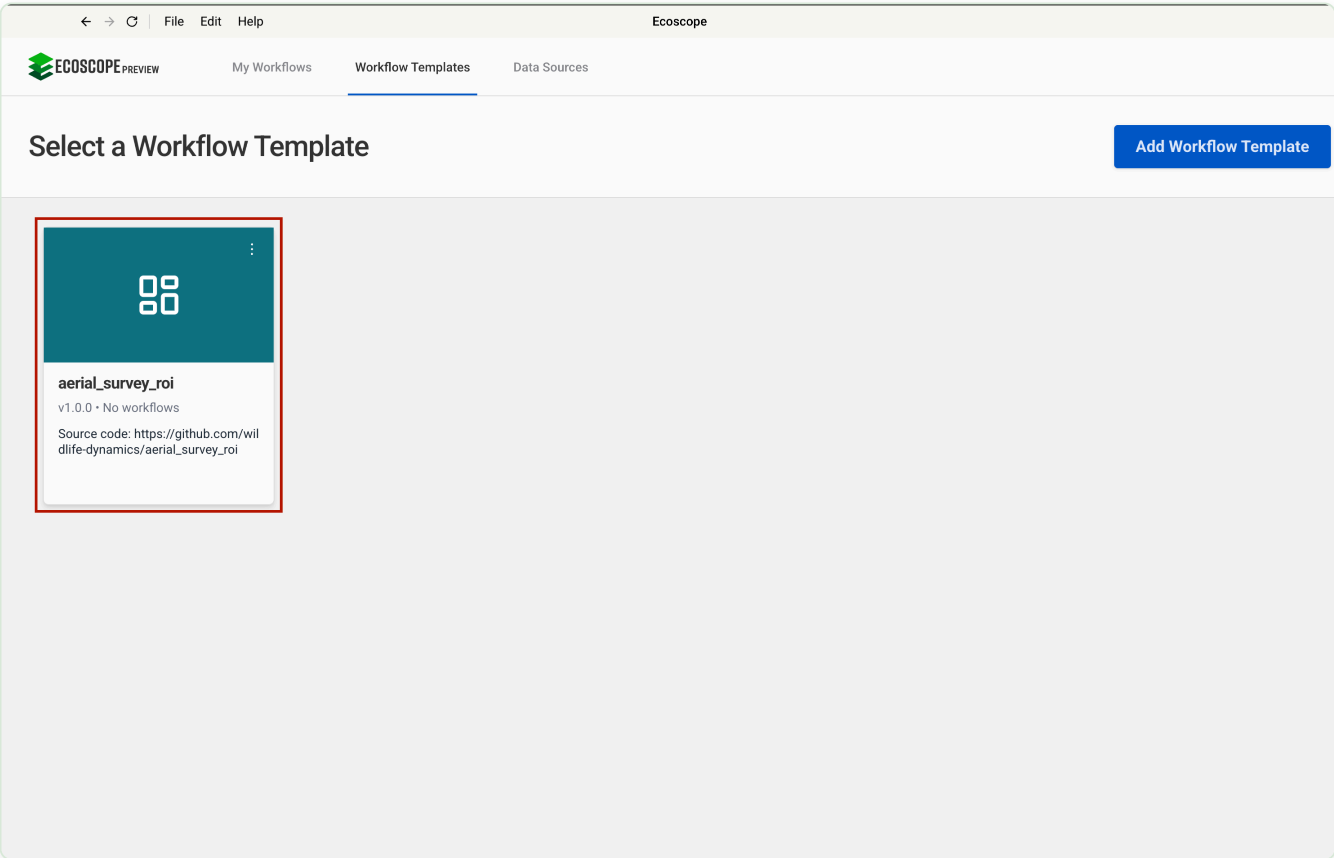
In the **Github Link** field, paste the following URL and click **Add Template**:

```
https://github.com/wildlife-dynamics/aerial_survey_roi.git
```



Step 2 — Select the Workflow to Configure

After the template is added, it appears as `aerial_survey_roi` under **Workflow Templates**. Click the card to open and configure it.



Step 3 — Configure Workflow Details

The **Configure New Workflow** page has a left-side navigation panel with four sections. Complete the first two:

Set Workflow Details

Provide a name and optional description to identify this workflow run. This label appears in the output dashboard.

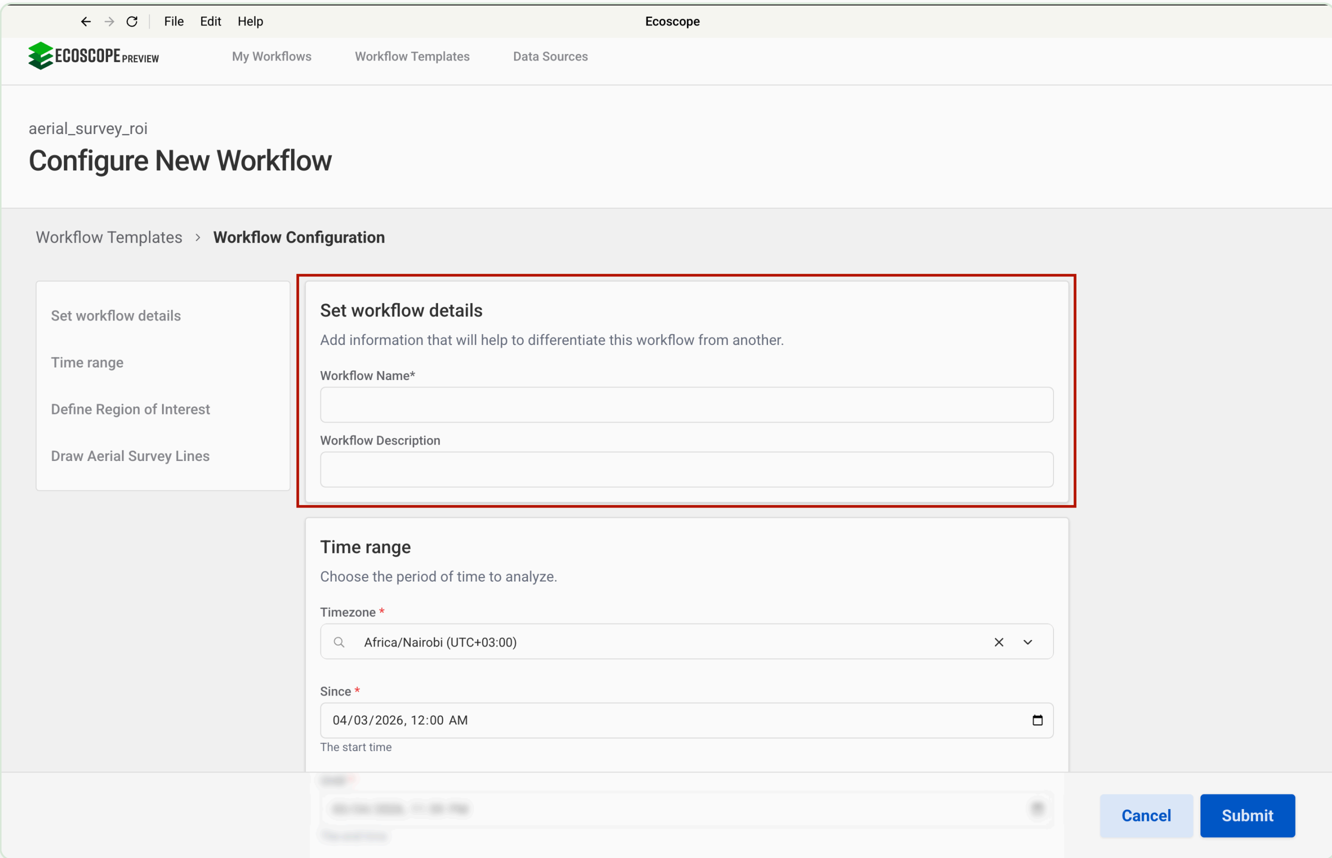
Field	Required	Example
Workflow Name	Yes	`Mara North Conservancy Aerial Survey`
Workflow Description	No	`Q1 2026 aerial transect survey`

Time Range

Select the timezone and start/end time for the analysis period.

Field	Example
Timezone	`Africa/Nairobi (UTC+03:00)`

Field	Example
Since (start)	`04/03/2026, 12:00 AM`
Until (end)	End of your survey window



Step 4 — Define ROI and Draw Aerial Survey Lines

Complete the final two sections in the left panel.

Define Region of Interest

Provide the boundary file for the area to be surveyed. Supported input methods:

Input Method	Example
Download from URL	`https://www.dropbox.com/.../mnc_conservancy.gpkg`
Local path	`/data/inputs/my_conservancy.gpkg`

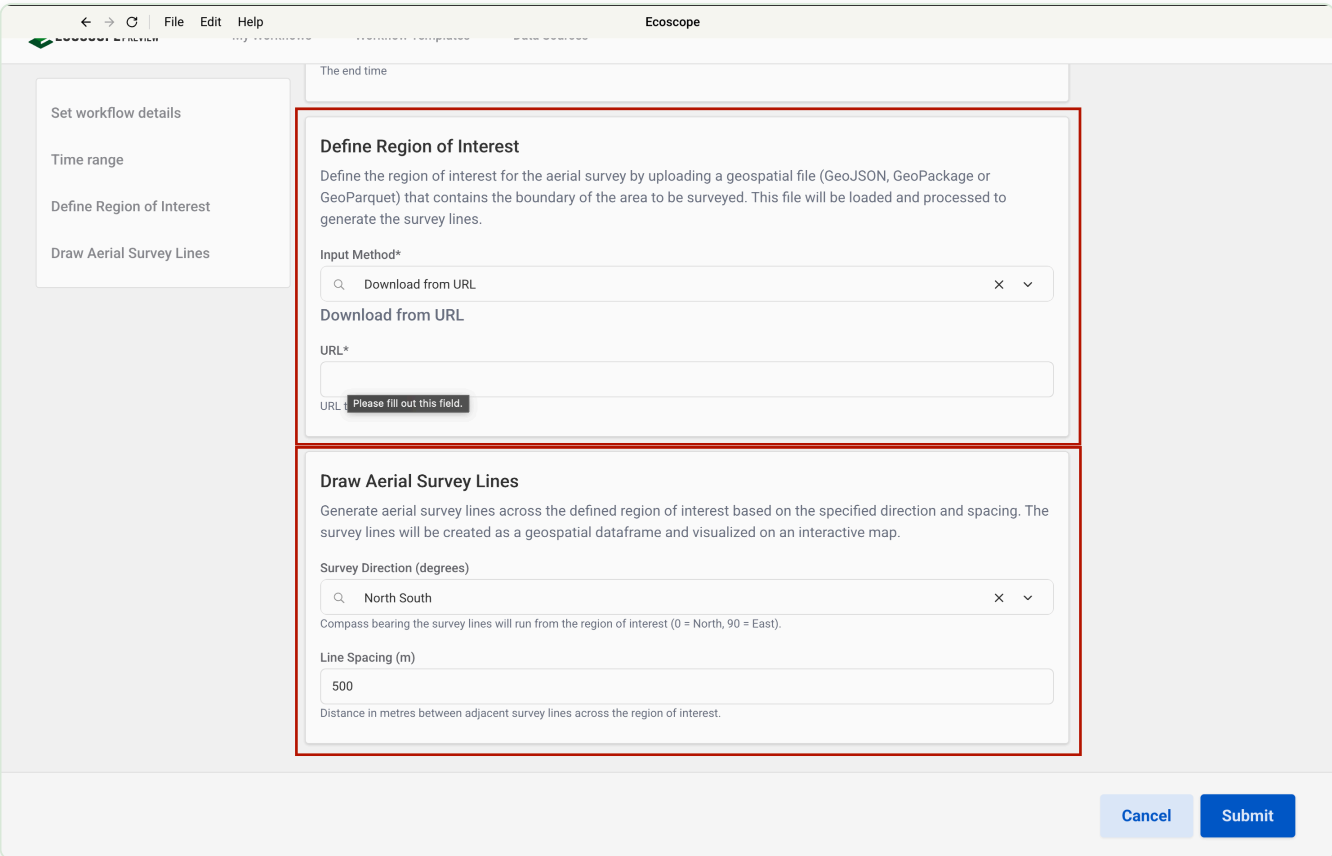
- Select **Download from URL** as the Input Method and paste a direct download link to your `.gpkg`, `.geojson`, or `.geoparquet` file.
- Any EPSG coordinate reference system is accepted — the workflow reprojects to EPSG:4326 automatically.

- **The file must contain polygon geometry.** Point or line files will cause the workflow to fail.

Draw Aerial Survey Lines

Configure how transect lines are generated across the ROI. Lines are automatically clipped to the polygon boundary.

Parameter	Description	Example
Survey Direction (Degrees)	Compass bearing / orientation of the transects	`North South`
Line Spacing (m)	Distance in metres between adjacent survey lines	`500`



When all sections are complete, click **Submit** in the bottom-right corner to run the workflow.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` :

File	Format	Description
`survey_lines_<hash>.geoparquet`	GeoParquet	Survey transect lines in cloud-native format (a content hash is appended to the filename to avoid

File	Format	Description
		overwriting previous runs)
`aerial_survey.html`	HTML	Interactive ecomap with ROI boundary and survey lines

The dashboard displays a map widget titled **"Aerial Survey Lines"** showing the ROI boundary (olive green fill, 25% opacity) and the generated survey transects (yellow lines).

Troubleshooting

Issue	Likely Cause	Fix
Workflow fails on ROI input	Input file contains non-polygon geometry	Ensure your file contains only polygon features
No lines generated	ROI too small relative to line spacing	Reduce the **Line Spacing** value
URL field validation error	URL is not a direct download link	Use a direct file URL, not a browser preview link